

ICT ACADEMY – CYBER SECURITY PROGRAM

Day 4 – Windows RDP, Compression Tools & Linux Filesystem with AWS EC2 Practical Lab

Learning Log / GitHub Documentation

Session Details

Participant	Akshaya Lad P
Day	Day 4
Date	29 September 2026
Hands-on Platform	Amazon Web Services (AWS)
Cloud Services Used	Amazon EC2 (Windows Server & Amazon Linux 2023)
Access Methods	RDP (Windows) and SSH (Linux)

Day 4 Focus

Learn how to access a Windows EC2 instance graphically using RDP, understand file compression/decompression tools on Windows and Linux, review Linux user and root management concepts, and explore the Linux root filesystem hierarchy. Apply these concepts hands-on by connecting to a Windows EC2 instance via RDP and to a Linux EC2 instance via SSH to run commands and inspect the filesystem.

1. Connecting to a Windows EC2 Instance (RDP)

Step	Action
Step 1	Log into the AWS account and open the EC2 console.
Step 2	Launch a new Windows Server EC2 instance.
Step 3	Select the instance, click Connect, and switch to the RDP client tab.
Step 4	Download the RDP shortcut file for the instance.
Step 5	Get the Windows (Administrator) password for the instance.

2. Decrypting the Windows Password

- Upload the instance's private key (.pem file) on the Get Windows Password screen.
- Click Decrypt Password, then copy and paste the decrypted password into Notepad for safekeeping.
- Copy the instance's public IP address and the Remote Desktop Connection username (Administrator).

3. Connecting via Remote Desktop

1. Open the downloaded RDP shortcut file.
2. Click Connect in the Remote Desktop Connection prompt.
3. When asked for a password, paste the decrypted password (Ctrl+V) and click OK.
4. Accept the certificate warning by clicking Yes.
5. A new Windows desktop opens inside the local machine – the remote Windows Server session.

4. Compression & Decompression Tools

Platform	Common Tools
Windows	ZIP, WinRAR, 7-Zip – used to compress and extract files and folders.
Linux	zip / gzip (gun zip), and advanced archive tools such as tar – used to compress, extract and manage archives from the command line.

5. Linux User & Root Management

Windows user types	Administrator user, Standard user, Normal user.
Linux user types	Root user, System user, Normal user.
Background processes	Linux services that run in the background are commonly called daemons.

6. Quick System Identification Commands

1. How to identify which Linux OS (distribution) is currently in use.
2. How to identify the kernel version currently running.
3. How to identify which shell is in use, and its version.

(history also shows how many commands have been run previously in the current session.)

7. Linux File System Overview

Directory	Purpose
/	File system root – the top of the entire Linux directory tree; everything starts here (cd /, ls /).
/bin	Essential user commands (ls, cp, mv, cat, mkdir) needed by the system to run.
/sbin	Essential system administration commands (e.g. fsck, shutdown, reboot).
/boot	Boot files used during the boot process, e.g. the vmlinuz kernel image.
/dev	Device files representing hardware and virtual devices, e.g. /dev/sda, /dev/null, /dev/random, /dev/tty.
/etc	System-wide configuration files, e.g. /etc/passwd, /etc/shadow, /etc/group, /etc/hosts, /etc/crontab.
/home	Personal home directories for regular users, e.g. /home/akshaya, /home/student.
/root	The root user's personal home directory (different from /, the file system root).
/lib	Shared libraries used by programs and system binaries.
/usr	User/system applications; contains a large portion of installed software and supporting files (/usr/bin, /usr/sbin, /usr/lib).
/var	Variable data that changes frequently, e.g. /var/log, /var/tmp, /var/lib.
/tmp	Temporary files; not persistent across reboots.
/proc	A virtual file system exposing live process, kernel, memory and CPU information; not stored on disk.
/media, /mnt	Mount points for removable and temporarily mounted storage.
/opt, /srv, /run, /sys	Optional/add-on software, service data, runtime data and kernel/device information respectively.

8. Practical Lab – AWS EC2 (Windows RDP & Linux SSH)

Goal: Access a Windows EC2 instance graphically using RDP, and separately connect to a Linux EC2 instance over SSH to practice core commands and explore the root filesystem.

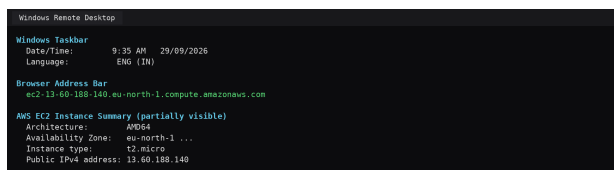
Lab Steps	
Step 1 – Launch instances	Launch a Windows Server EC2 instance (for RDP) and an Amazon Linux 2023 EC2 instance (for SSH).
Step 2 – RDP setup	In the EC2 console, select the Windows instance, click Connect, switch to the RDP tab, and download the RDP shortcut file.
Step 3 – Decrypt password	Upload the private key (.pem) to decrypt the Administrator password, then copy it to Notepad.
Step 4 – Connect via RDP	Open the RDP shortcut file, click Connect, paste the password, accept the certificate, and reach the remote Windows desktop.
Step 5 – SSH into Linux	Connect to the Linux EC2 instance as ec2-user using the private key pair.
Step 6 – Practice commands	Run pwd, touch and history to explore the home directory and confirm command history tracking (including a "history" typo producing "command not found").
Step 7 – Download software	Reconnect over SSH and use sudo wget to download Apache Tomcat directly onto the instance, then verify with ls and ls -ltr.
Step 8 – Explore root filesystem	Run cd / followed by ls -ltr to inspect the top-level Linux directories (bin, boot, dev, etc, home, proc, var and others).

SSH Command Used in the Practical

ssh -i "path-to-your-key.pem" ec2-user@YOUR_EC2_PUBLIC_IP
Important: Never upload the .pem private key, decrypted passwords, access keys or other secrets to GitHub.

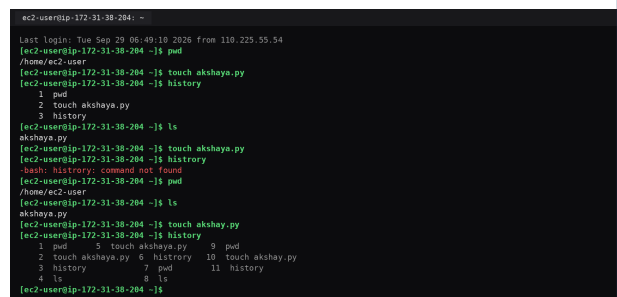
Practical Evidence – AWS EC2 (Windows RDP & Linux SSH)

1. Windows RDP Session Info



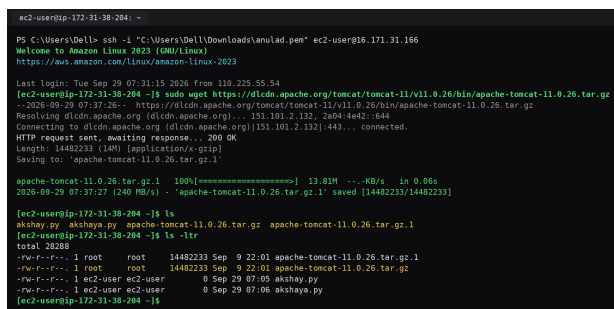
Windows taskbar date/time and the EC2 instance's public DNS address bar, confirming the Remote Desktop session on the Windows Server instance.

2. Linux Shell – pwd / touch / history



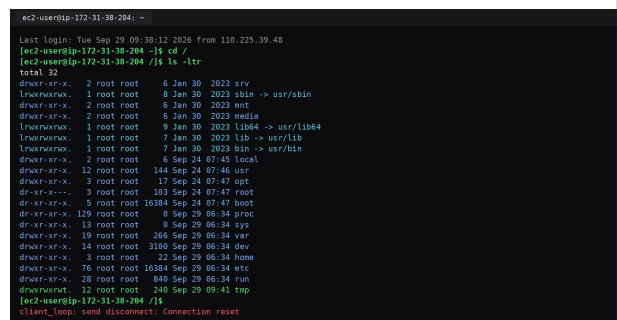
pwd confirms the home directory; touch creates files; history lists commands run so far, including a mistyped "history" command.

3. SSH Login & Downloading Apache Tomcat



SSH login to Amazon Linux 2023, followed by sudo wget downloading Apache Tomcat, then ls -ltr confirming the saved files.

4. Exploring the Root Filesystem



cd / followed by ls -ltr, listing the top-level Linux directories such as bin, boot, dev, etc, home, proc, var and tmp.

9. What I Learned on Day 4

1. Learned how to launch and access a Windows EC2 instance graphically using RDP.
2. Understood how to decrypt the Windows Administrator password using the instance's private key.
3. Practiced the full RDP connection flow: download shortcut, connect, paste password, accept certificate.
4. Reviewed compression and decompression tools on Windows (ZIP, WinRAR, 7-Zip) and Linux (zip, gzip, tar).
5. Learned the difference between Windows and Linux user types, and what a daemon is.
6. Learned how to check the Linux distribution, kernel version and shell version on a system.
7. Studied the Linux root filesystem hierarchy and the purpose of each top-level directory.
8. Practiced pwd, touch and history on a live Amazon Linux 2023 EC2 instance.
9. Used sudo wget to download and verify software (Apache Tomcat) directly on the server.
10. Explored the real root filesystem on EC2 with cd / and ls -ltr and matched it against the theory.